corresponding to all other states—all states other than i and j become transient states. The set of all states other than states i and j forms an *open* subset of states and the results of Theorem 6.1.9 can be applied. In other words, the problem posed can be solved by turning states i and j into absorbing states and determining the probability that, starting in some nonabsorbing state k, the Markov chain enters absorbing state i before it enters absorbing state j.

Assume that there are a total of n states and that these are ordered as follows: the set of $(n-2)$ states which does not contain i or j, but which does contain state k, appears first, followed by state i and finally state j. The probability that state i is entered before state j is given by the elements of the vector formed as the product of S and a vector, denoted v_{n-1}, whose components are the first $n-2$ elements of column $n-1$ of P, i.e., the elements that define the conditional probabilities of entering state i given that the Markov chain is in state k, $k = 1, 2, \ldots, n-2$. Similarly, the probability that state j is reached before state i is computed from the elements of the vector obtained when S is multiplied by a vector, denoted v_n, whose components are the first $n-2$ elements of the last column of P.

Example 9.13 Consider the six-state Markov chain whose transition probability matrix is given by

$$P = \left(\begin{array}{cccc|cc} .25 & .25 & & & .50 & \\ .50 & & .50 & & & \\ & .50 & & .50 & & \\ & & .50 & & & .50 \\ \hline & & & .50 & & .50 \\ & & & & .50 & .50 \end{array}\right).$$

We have

$$S = \begin{pmatrix} .75 & -.25 & & \\ -.50 & 1.0 & -.50 & \\ & -.50 & 1.0 & -.50 \\ & & -.50 & 1.0 \end{pmatrix}^{-1} = \frac{2}{9}\begin{pmatrix} 8.0 & 3.0 & 2.0 & 1.0 \\ 6.0 & 9.0 & 6.0 & 3.0 \\ 4.0 & 6.0 & 10.0 & 5.0 \\ 2.0 & 3.0 & 5.0 & 7.0 \end{pmatrix}.$$

With

$$v_5 = \big(0.5\ 0.0\ 0.0\ 0.0\big)^T \quad \text{and} \quad v_6 = \big(0.0\ 0.0\ 0.0\ 0.5\big)^T,$$

we obtain

$$Sv_5 = \big(0.8889\ 0.6667\ 0.4444\ 0.2222\big)^T \quad \text{and} \quad Sv_6 = \big(0.1111\ 0.3333\ 0.5556\ 0.7778\big)^T.$$

Thus, for example, the probability that state $i = 5$ is reached before state $j = 6$ given state $k = 1$ as the starting state is 0.8889. The probability that state j is reached before state i is only 0.1111. Notice that combining v_5 and v_6 into a single 4×2 matrix gives what we previously referred to as the matrix B, so what we are actually computing is just $SB = A$, the matrix of absorption probabilities.

9.7 Random Walk Problems

Picture a drunken man trying to walk along an imagined straight line on his way home from the pub. Sometimes he steps to the right of his imagined straight line and sometimes he steps to the left of it. This colorful scenario leads to an important class of Markov chain problems, that of the random walk. The states of the Markov chain are the integers $0, \pm 1, \pm 2, \ldots$ (the drunkard's straight line) and the only transitions from any state i are to neighboring states $i+1$ (a step to the right) with probability p and $i-1$ (a step to the left) with probability $q = 1 - p$. The state transition diagram is shown in Figure 9.12 and the process begins in state 0. When $p = 1/2$, the process is said to be a *symmetric* random walk.

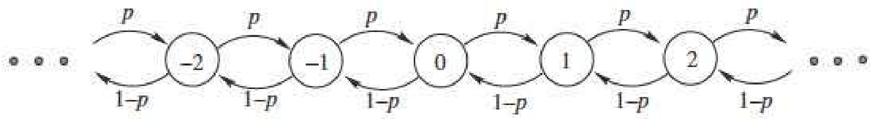

Figure 9.12. State transition diagram for a random walk on the integers.

There are many variations on this basic theme. The case in which the state space is finite is sometimes called the "gambler's ruin" problem. A gambler starts with a certain amount of money, say i dollars (in terms of the Markov chain, the process begins in state $i > 0$) and on each play the gambler can either win a dollar with probability p or lose a dollar with probability $1 - p$. The interest is in determining whether the gambler is ruined, i.e., loses all his money (the Markov chain moves to state 0, which is taken to be an absorbing state) or wins a fortune (the Markov chain moves into absorbing state $N > i$, where N is large). Other variations are based on a random walk that is infinite on one side only, the state space is taken to be the set of nonnegative integers, and the Markov chain begins in state 0. Random walk problems can also be defined on more than one dimension. Two-dimensional random walks take place on a grid that can be infinite and cover the entire plane; infinite but span only the upper right quadrant; or finite. In these cases, each state communicates with at most its four neighbors to the north, south, east, and west. The process is symmetric when each transition probability is equal to 1/4. Random walks on higher dimensions are readily envisaged.

We begin by considering random walk problems in which the Markov chain is irreducible, when there is a path from any state to any other state. In such circumstances we know that all of the states are either positive recurrent or all are null recurrent or all are transient. Our interest is in finding out which applies. In the original one-dimensional random walk on the integers $0, \pm 1, \pm 2, \ldots$, the states are all transient except in the symmetric case, $p = 1/2$. The proof of this is outlined in the exercises. The same result holds in two dimensions, but in three dimensions or higher, all the states are transient, even in the symmetric case. We now analyze the case of a random walk on the nonnegative integers and leave some of the other cases to the exercises. The following two theorems (stated without proof) are useful in this respect.

Theorem 9.7.1 *Let P be the single-step transition probability matrix of an irreducible Markov chain. Then all the states of this Markov chain are positive recurrent if and only if the system of linear equations*

$$z = zP,$$

in which z is a row vector, has a solution with $\sum_{\text{all } j} z_j = 1$. If such a solution exists, then we must have $z_j > 0$ for all j and there are no other solutions.

We shall see momentarily that any vector $z \neq 0$ which satisfies $z = zP$, not necessarily one for which *all* the components of z are strictly positive, is called a *stationary distribution* of the Markov chain. The second theorem that we shall need is the following.

Theorem 9.7.2 *Let P be the single-step transition probability matrix of an irreducible Markov chain. Let P' be the matrix obtained from P by deleting row k and column k. Then all states are recurrent if and only if the solution y of*

$$y = P'y, \quad 0 \leq y_i \leq 1,$$

is the zero vector, i.e., a column vector whose components are all equal to zero.

Consider the infinite (denumerable) Markov chain whose transition probability matrix is given by

$$P = \begin{pmatrix} 0 & 1 & 0 & \cdots & & \\ q & 0 & p & 0 & \cdots & \\ 0 & q & 0 & p & 0 & \cdots \\ 0 & 0 & q & 0 & p & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$

where p is a positive probability and $q = 1 - p$. Observe that every time the Markov chain reaches state 0, it must leave it again at the next time step. State 0 is said to constitute a *reflecting barrier*. In other applications it may be more appropriate to set $p_{11} = 1 - p$ and $p_{12} = p$, called a *Bernoulli barrier*, or $p_{11} = 1$ and $p_{12} = 0$, called an *absorbing barrier*. We pursue our analysis with the reflecting barrier option. As we pointed out previously, since every state can reach every other state, the Markov chain is irreducible and hence all the states are positive recurrent or all the states are null recurrent or all the states are transient. Notice also that a return to any state is possible only in a number of steps that is a multiple of 2. Thus the Markov chain is periodic with period equal to 2. We now wish to use Theorems 9.7.1 and 9.7.2 to classify the states of this chain. Consider the system of equations $z = zP$.

$$(z_0\ z_1\ z_2 \ldots) = (z_0\ z_1\ z_2 \ldots) \begin{pmatrix} 0 & 1 & 0 & \cdots & & \\ q & 0 & p & 0 & \cdots & \\ 0 & q & 0 & p & 0 & \cdots \\ 0 & 0 & q & 0 & p & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Taking these equations one at a time, we find

$$z_0 = z_1 q \implies z_1 = \frac{1}{q} z_0,$$

$$z_1 = z_0 + q z_2 \implies z_2 = \frac{1}{q}(z_1 - z_0) = \frac{1}{q}(1-q) z_1 = \left(\frac{p}{q}\right) z_1 = \frac{1}{q}\left(\frac{p}{q}\right) z_0.$$

All subsequent equations are of the form

$$z_j = p z_{j-1} + q z_{j+1} \quad \text{for} \quad j \geq 2$$

and have the solution

$$z_{j+1} = \left(\frac{p}{q}\right) z_j,$$

which may be proven by induction as follows. Base clause, $j = 2$:

$$z_2 = p z_1 + q z_3 \implies z_3 = \frac{1}{q}(z_2 - p z_1) = \frac{1}{q} z_2 - z_2 = \left(\frac{p}{q}\right) z_2.$$

We now assume this solution to hold for j and prove it true for $j + 1$. From

$$z_j = p z_{j-1} + q z_{j+1}$$

we obtain

$$z_{j+1} = \frac{1}{q}(z_j - p z_{j-1}) = \left(\frac{1}{q}\right) z_j - \left(\frac{p}{q}\right) z_{j-1} = (1/q - 1) z_j = \left(\frac{p}{q}\right) z_j$$

which completes the proof. This solution allows us to write any component in terms of z_0. We have

$$z_j = \left(\frac{p}{q}\right) z_{j-1} = \left(\frac{p}{q}\right)^2 z_{j-2} = \cdots = \left(\frac{p}{q}\right)^{j-1} z_1 = \frac{1}{q}\left(\frac{p}{q}\right)^{j-1} z_0, \quad j \geq 1$$

and summing over all z_j, we find

$$\sum_{j=0}^{\infty} z_j = z_0 \left[1 + \frac{1}{q}\sum_{j=1}^{\infty}\left(\frac{p}{q}\right)^{j-1}\right].$$

Notice that the summation inside the square bracket is finite if and only if $p < q$ in which case

$$\sum_{j=1}^{\infty}\left(\frac{p}{q}\right)^{j-1} = \sum_{j=0}^{\infty}\left(\frac{p}{q}\right)^{j} = \frac{1}{1-p/q} \quad \text{(iff } p < q).$$

The sum of all z_j being equal to 1 implies that

$$1 = z_0\left[1 + \frac{1}{q}\left(\frac{1}{1-p/q}\right)\right].$$

Simplifying the term inside the square brackets, we obtain

$$1 = z_0\left[1 + \frac{1}{q-p}\right] = z_0\left[\frac{q-p+1}{q-p}\right] = z_0\left[\frac{2q}{q-p}\right]$$

and hence

$$z_0 = \frac{q-p}{2q} = \frac{1}{2}\left(1 - \frac{p}{q}\right) \quad \text{for } p < q.$$

The remaining z_j are obtained as

$$z_j = \frac{1}{q}\left(\frac{p}{q}\right)^{j-1} z_0 = \frac{1}{2q}\left(1 - \frac{p}{q}\right)\left(\frac{p}{q}\right)^{j-1}.$$

This solution satisfies the conditions of Theorem 9.7.1 (it exists and its components sum to 1) and hence all states are positive recurrent (when $p < q$). We also see that the second part of this theorem is true (all z_j are strictly positive). Finally, this theorem tells us that there is no other solution.

We now examine the other possibilities, namely, $p = q$ and $p > q$. Under these conditions, either all states are null recurrent or else all states are transient (from Theorems 9.5.1 and 9.7.1). We shall now use Theorem 9.7.2 to determine which case holds. Let the matrix P' be the matrix obtained from P when the first row and column of P is removed and let us consider the system of equations $y = P'y$. We have

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 & p & 0 & 0 & \cdots \\ q & 0 & p & 0 & \cdots \\ 0 & q & 0 & p & \cdots \\ \vdots & \vdots & \ddots & \ddots & \ddots \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \end{pmatrix}.$$

Again taking these equations one at a time, we have

$$y_1 = py_2,$$
$$y_2 = qy_1 + py_3,$$

and in general

$$y_j = qy_{j-1} + py_{j+1}.$$

Writing the left-hand side as $py_j + qy_j$, we have

$$py_j + qy_j = qy_{j-1} + py_{j+1} \quad \text{for } j \geq 2,$$

which yields

$$p(y_{j+1} - y_j) = q(y_j - y_{j-1}) \quad \text{for } j \geq 2. \tag{9.18}$$

When $j = 1$, a similar rearrangement gives

$$p(y_2 - y_1) = qy_1.$$

It follows from Equation (9.18) that

$$\begin{aligned} y_{j+1} - y_j &= \left(\frac{q}{p}\right)(y_j - y_{j-1}) = \cdots = \left(\frac{q}{p}\right)^{j-1}(y_2 - y_1) \\ &= \left(\frac{q}{p}\right)^{j-1}\left(\frac{q}{p}\right) y_1 = \left(\frac{q}{p}\right)^{j} y_1 \quad \text{for } j \geq 1. \end{aligned}$$

Thus

$$y_{j+1} - y_j = \left(\frac{q}{p}\right)^{j} y_1 \quad \text{for } j \geq 1$$

and

$$\begin{aligned} y_{j+1} &= (y_{j+1} - y_j) + (y_j - y_{j-1}) + \cdots + (y_2 - y_1) + y_1 \\ &= \left(\frac{q}{p}\right)^{j} y_1 + \left(\frac{q}{p}\right)^{j-1} y_1 + \cdots + \left(\frac{q}{p}\right)^{1} y_1 + y_1 \\ &= \left[1 + \left(\frac{q}{p}\right) + \left(\frac{q}{p}\right)^{2} + \cdots + \left(\frac{q}{p}\right)^{j}\right] y_1. \end{aligned}$$

Consider now the case in which $p = q$. We obtain

$$y_j = jy_1 \quad \text{for } j \geq 1.$$

In order to satisfy the conditions of Theorem 9.7.2, we need to have $0 \leq y_j \leq 1$ for all j. Therefore the only possibility in this case ($p = q$) is that $y_1 = 0$, in which case $y_j = 0$ for all j. Since the only solution is $y = 0$, we may conclude from Theorem 9.7.2 that all the states are *recurrent* states and since we know that they are not *positive recurrent*, they must be null recurrent.

Finally, consider the case in which $p > q$. In this case, q/p is a fraction and the summation converges. We have

$$y_j = \left[1 + \left(\frac{q}{p}\right) + \left(\frac{q}{p}\right)^{2} + \cdots + \left(\frac{q}{p}\right)^{j-1}\right] y_1 = \sum_{k=0}^{j-1} \left(\frac{q}{p}\right)^{k} y_1 = \frac{1 - (q/p)^j}{1 - (q/p)} y_1.$$

It is apparent that if we now set

$$y_1 = 1 - \frac{q}{p}$$

we obtain a value of y_1 that satisfies $0 \leq y_1 \leq 1$ and, furthermore, we also obtain

$$y_j = 1 - \left(\frac{q}{p}\right)^{j},$$

which also satisfies $0 \leq y_j \leq 1$ for all $j \geq 1$. From Theorem 9.7.2 we may now conclude that all the states are transient when $p > q$.

Let us now consider a different random walk, the gambler's ruin problem. The gambler begins with i dollars (in state i) and on each play either wins a dollar with probability p or loses a dollar with probability $q = 1 - p$. If X_n is the amount of money he has after playing n times, then $\{X_n, n = 0, 1, 2, \ldots\}$ is a Markov chain and the transition probability matrix is given by

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & & \cdots & & 0 \\ q & 0 & p & 0 & & \cdots & & 0 \\ 0 & q & 0 & p & & \cdots & & 0 \\ & & & & & & & \\ \vdots & & \ddots & \ddots & \ddots & & & \vdots \\ & & & & & & & \\ 0 & & \cdots & & q & 0 & p & 0 \\ 0 & & \cdots & & 0 & q & 0 & p \\ 0 & & \cdots & & 0 & 0 & 0 & 1 \end{pmatrix}.$$

We would like to find the probability that the gambler will eventually make his fortune by arriving in state N given that he starts in state i. Let x_i be this probability. We immediately have that $x_0 = 0$ and $x_N = 1$, since the gambler cannot start with zero dollars and win N dollars, while if he starts with N dollars he has already made his fortune. Given that the gambler starts with i dollars, after the first play he has $i + 1$ dollars with probability p, and $i - 1$ dollars with probability q. It follows that the probability of ever reaching state N from state i is the same as the probability of reaching N beginning in $i + 1$ with probability p plus the probability of reaching state N beginning in state $i - 1$ with probability q. In other words,

$$x_i = px_{i+1} + qx_{i-1}.$$

This is the equation that will allow us to solve our problem. It holds for all $1 \leq i \leq N - 1$. Be aware that it does not result from the multiplication of a vector by the transition probability matrix, but rather by conditioning on the result of the first play. Writing this equation as

$$x_i = px_i + qx_i = px_{i+1} + qx_{i-1}$$

allows us to derive the recurrence relation

$$x_{i+1} - x_i = \frac{q}{p}(x_i - x_{i-1}), \quad i = 1, 2, \ldots, N - 1.$$

Now, using the fact that $x_0 = 0$, we have

$$\begin{aligned} x_2 - x_1 &= \frac{q}{p}(x_1 - x_0) = \frac{q}{p}x_1, \\ x_3 - x_2 &= \frac{q}{p}(x_2 - x_1) = \left(\frac{q}{p}\right)^2 x_1, \\ &\vdots \\ x_i - x_{i-1} &= \frac{q}{p}(x_{i-1} - x_{i-2}) = \left(\frac{q}{p}\right)^{i-1} x_1, \qquad i = 2, 3, \ldots, N. \end{aligned}$$

Adding these equations, we find

$$x_i - x_1 = \left[\left(\frac{q}{p}\right) + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1}\right] x_1,$$

i.e.,

$$x_i = \left[1 + \left(\frac{q}{p}\right) + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1}\right] x_1 = \sum_{k=0}^{i-1} \left(\frac{q}{p}\right)^k x_1.$$

We must consider the two possible cases for this summation, $p \neq q$ and $p = q = 1/2$. When $p \neq q$, then

$$x_i = \frac{1-(q/p)^i}{1-q/p} x_1 \quad \text{and in particular} \quad \frac{1-(q/p)^N}{1-q/p} x_1 = x_N = 1.$$

This allows us to compute x_1 as

$$x_1 = \frac{1-q/p}{1-(q/p)^N},$$

and from this, all the remaining values of x_i, $i = 2, 3, \ldots, N-1$, can be found. Collecting these results together, we have

$$x_i = \frac{1-(q/p)^i}{1-(q/p)} \times \frac{1-(q/p)}{1-(q/p)^N} = \frac{1-(q/p)^i}{1-(q/p)^N}, \quad i = 1, 2, \ldots N \quad \text{and} \quad p \neq q.$$

Notice the limits as N tends to infinity:

$$\lim_{N\to\infty} x_i = \begin{cases} 1-(q/p)^i, & p > 1/2, \\ 0, & p < 1/2. \end{cases}$$

This means that, when the game favors the gambler ($p > 1/2$), there is a positive probability that the gambler will make his fortune. However, when the game favors the house, then the gambler is sure to lose all his money. This same sad result holds when the game favors neither the house nor the gambler ($p = 1/2$). We leave it as an exercise for the reader to show that, when $p = q = 1/2$, then $x_i = i/N$, for $i = 1, 2, \ldots, N$, and hence x_i approaches zero as $N \to \infty$.

Example 9.14 Billy and his brother Gerard play marbles. At each game Billy has a 60% chance of winning (perhaps he cheats) while Gerard has only a 40% chance of winning. If Billy starts with 16 marbles and Gerard with 20, what is the probability that Billy ends up with all the marbles? Suppose Billy starts with only 4 marbles (and Gerard with 20), what is the probability that Gerard ends up with all the marbles?

We shall use the following equation generated during the gambler's ruin problem:

$$x_i = \frac{1-(q/p)^i}{1-(q/p)^N}, \quad i = 1, 2, \ldots, N \quad \text{and} \quad p \neq q.$$

Substituting in the values $p = 0.6$, $i = 16$, and $N = 36$, we have

$$x_{16} = \frac{1-(.4/.6)^{16}}{1-(.4/.6)^{36}} = 0.998478.$$

To answer the second part, the probability that Gerard, in a more advantageous initial setup, wins all the marbles, we first find x_4, the probability that Billy wins them all. We have

$$x_4 = \frac{1-(.4/.6)^4}{1-(.4/.6)^{24}} = 0.802517.$$

So the probability that Gerard takes all of Billy's marbles is only $1 - 0.802517 = 0.197483$.